CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 12: **Reproduction: How Life Continues** | Quick revision, one page

Asexual reproduction

- Single parent, no gametes, offspring identical to the parent.
- Binary fission: *Amoeba*, bacteria. Budding: yeast, *Hydra*. Spore formation: bread mould (*Rhizopus*), ferns. Fragmentation: *Spirogyra*. Regeneration: *Planaria*, *Hydra*.
- Vegetative propagation: cutting (rose), layering (jasmine), grafting (mango), tubers (potato), bulbs (onion). Tissue culture: new plants from small pieces of tissue.

Sexual reproduction

- Two parents; fusion of male and female gametes gives a zygote; offspring show variation.
- Number of cells after n divisions of a zygote $= 2^n$ (equal divisions).
- Human chromosomes: 23 pairs (46); gametes have 23; sex chromosomes XX (female) and XY (male).

Flowering plants

- Stamen (anther with pollen) is male; pistil (stigma, style, ovary with ovules) is female.
- Pollination $\rightarrow$ pollen germinates, pollen tube $\rightarrow$ fertilisation $\rightarrow$ zygote $\rightarrow$ embryo; ovule $\rightarrow$ seed, ovary $\rightarrow$ fruit.
- Self-pollination: same flower or same plant; cross-pollination: different plant, by wind, water, insects, birds. Seed dispersal by wind, water, animals, explosion.

Human reproduction

- Male: testes (make sperm; in scrotum), vas deferens, urethra, penis. Female: ovaries (make eggs), fallopian tubes (site of fertilisation), uterus (implantation), vagina.
- Fertilisation $\rightarrow$ zygote $\rightarrow$ embryo $\rightarrow$ foetus; placenta supplies nutrients and oxygen; pregnancy about 9 months (about 40 weeks).
- Menstruation: shedding of the uterine lining about every 28 days when the egg is not fertilised.

Fertilisation in animals

- External fertilisation (outside the body, in water): fish, frogs; very many eggs. Internal fertilisation: birds, mammals, humans; fewer eggs, better protection.
- Identical twins: one zygote splits, same sex; fraternal twins: two eggs fertilised separately, may differ in sex.

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